Independent reproduction check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Resolve metric and scoring disagreements from artifacts without inventing a successful reproduction.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

Peer review and response

> **My remembered idea, example and correction**

Writing the methods section

> **My remembered idea, example and correction**

Robustness across conditions

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Independent reproduction uses the recorded materials to recover the stated computation with assistance and mismatches disclosed. Main paper calls the metric MSE while the script averages absolute errors, MAE. Formulas differ; preserve original artifacts and document correction/rerun. Actual original residuals are absent.

  An explicitly separate fixture with predictions 1, 3 and targets 1, 1 has residuals 0, 2: MAE one, MSE two. It demonstrates the contract without reconstructing the missing run. Separate scoring picture has original 5/6, rerun 4/6 and Q3 as the only disputed case, with no new model outputs. Inspect saved output, reference label and rubric; do not pick the higher score or average incompatible judgments.

  If Q3's saved file is missing, exact verification remains unresolved. Different split membership prevents an exact same-materials check. Record data/code/environment versions and assistance. Same-data reproduction is not new-sample replication, and the course's local fixture is not an independent full-study check.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  An explicitly separate fixture with predictions 1, 3 and targets 1, 1 has residuals 0, 2: MAE one, MSE two. It demonstrates the contract without reconstructing the missing run. Separate scoring picture has original 5/6, rerun 4/6 and Q3 as the only disputed case, with no new model outputs. Inspect saved output, reference label and rubric; do not pick the higher score or average incompatible judgments.

  If Q3's saved file is missing, exact verification remains unresolved. Different split membership prevents an exact same-materials check. Record data/code/environment versions and assistance. Same-data reproduction is not new-sample replication, and the course's local fixture is not an independent full-study check.

- Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Main metric audit	Supported statement
Named metric	MSE averages squared residuals
Implemented metric	MAE averages absolute residuals
Contract mismatch	Different computation; inspect/correct script or report with provenance
Actual original discrepancy size	Unknown; no main residual values supplied
Distinct fixture residuals	0 and 2
Fixture MAE	(0+2)/2 = 1

Main metric audit	Supported statement
Fixture MSE	$(0^2+2^2)/2 = 2$
Fixture purpose	Demonstrates possible difference, not the missing original run
After correction	Preserve old artifacts/results and record a new documented rerun

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Independent reproduction uses the recorded materials to recover the stated computation with assistance and mismatches disclosed. Main paper calls the metric MSE while the script averages absolute errors, MAE. Formulas differ; preserve original artifacts and document correction/rerun. Actual original residuals are absent.

  An explicitly separate fixture with predictions 1, 3 and targets 1, 1 has residuals 0, 2: MAE one, MSE two. It demonstrates the contract without reconstructing the missing run. Separate scoring picture has original 5/6, rerun 4/6 and Q3 as the only disputed case, with no new model outputs. Inspect saved output, reference label and rubric; do not pick the higher score or average incompatible judgments.

  If Q3's saved file is missing, exact verification remains unresolved. Different split membership prevents an exact same-materials check. Record data/code/environment versions and assistance. Same-data reproduction is not new-sample replication, and the course's local fixture is not an independent full-study check.

- A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Independent reproduction uses the recorded materials to recover the stated computation with assistance and mismatches disclosed. Main paper calls the metric MSE while the script averages absolute errors, MAE. Formulas differ; preserve original artifacts and document correction/rerun. Actual original residuals are absent.

  An explicitly separate fixture with predictions 1, 3 and targets 1, 1 has residuals 0, 2: MAE one, MSE two. It demonstrates the contract without reconstructing the missing run. Separate scoring picture has original 5/6, rerun 4/6 and Q3 as the only disputed case, with no new model outputs. Inspect saved output, reference label and rubric; do not pick the higher score or average incompatible judgments.

  If Q3's saved file is missing, exact verification remains unresolved. Different split membership prevents an exact same-materials check. Record data/code/environment versions and assistance. Same-data reproduction is not new-sample replication, and the course's local fixture is not an independent full-study check.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Reproducer uses different validation split because no split IDs were provided. Diagnose.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Reproducer uses different validation split because no split IDs were provided. Diagnose.

  Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Independent reproduction check. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Independent reproduction check. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Reproducer uses different validation split because no split IDs were provided. Diagnose.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

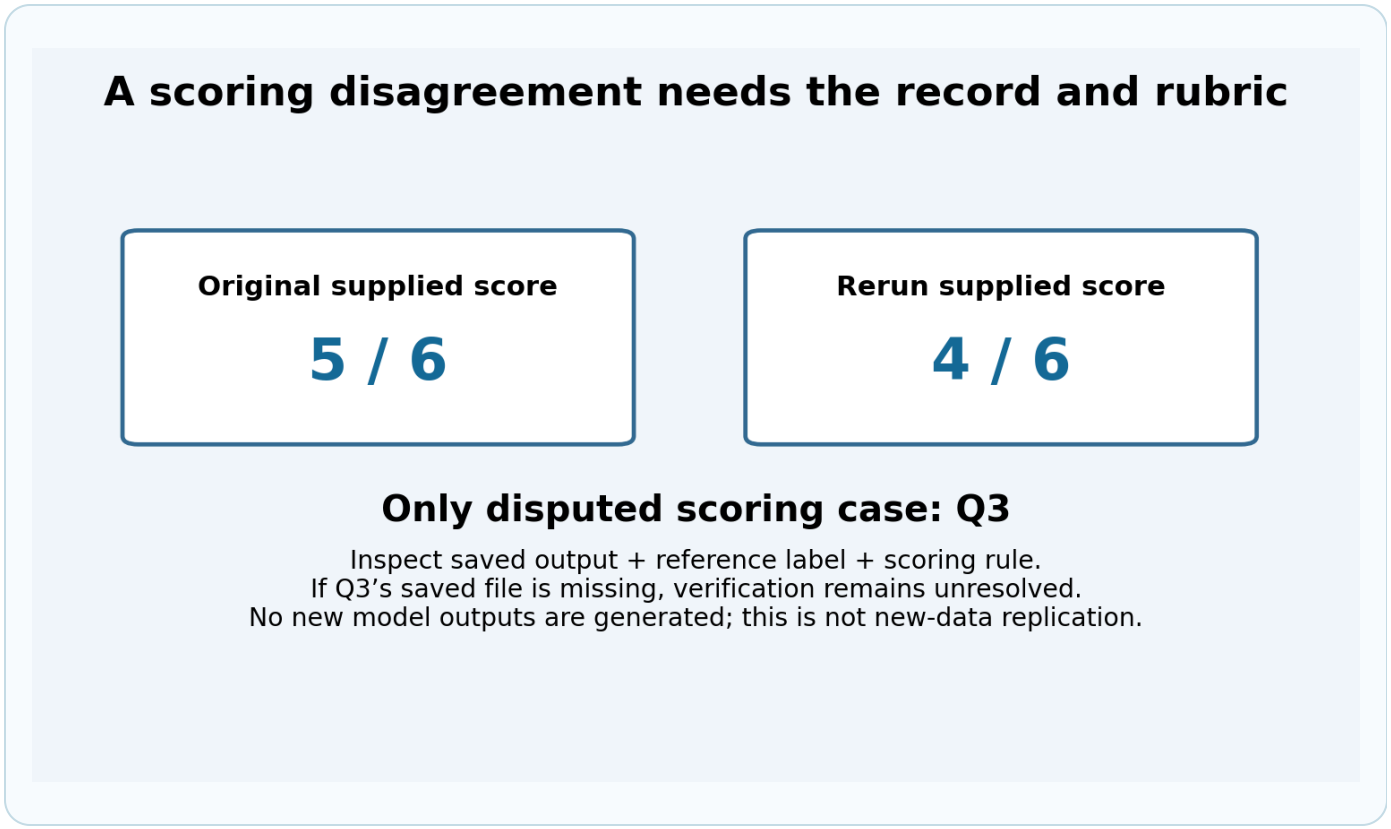
- Original correct count is 5/6. Rerun is 4/6. Case Q3 is the only scoring disagreement. No new model **output**s are generated.

- Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/33

Independent reproduction check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare exact artefacts

Verify data and rubric versions before changing the result.



2. Locate disagreement

The score difference is concentrated in Q3.



3. Resolve by the rubric

Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

CHANGE ONE THING

Q3's saved file is missing from the archive.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.
- Which labelled step matches this detail? Inspect Q3's saved answer and rule; document whichever scoring decision is justified.
- 1. Resolve by the rubric
- 2. Compare exact artefacts
- 3. Locate disagreement
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Resolve by the rubric Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Changed-case question and optional hint

- Changed-case practice: Q3's saved file is missing from the archive. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

Repair and retain the record

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer’s concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Independent reproduction check: the case

AI ACADEMY / AI-7/33 · 01 · LOOK AT THE FACTS

Independent reproduction check: the case

The situation

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

Supplied evidence

Original correct count is 5/6. Rerun is 4/6. Case Q3 is the only scoring disagreement. No new model outputs are generated.

Explain independent reproduction check using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

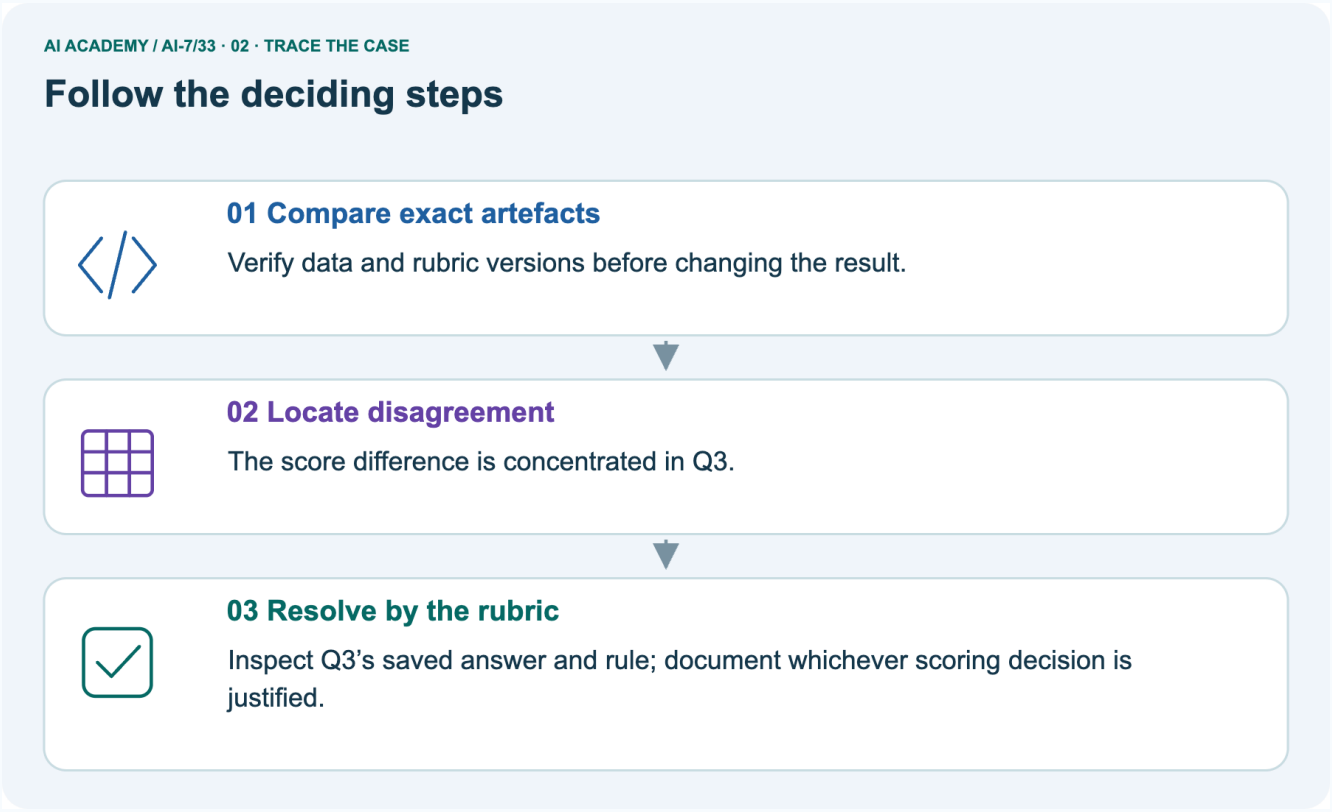
An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score. Supplied facts: Original correct count is 5/6. Rerun is 4/6.

Case Q3 is the only scoring disagreement. No new model **output**s are generated.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



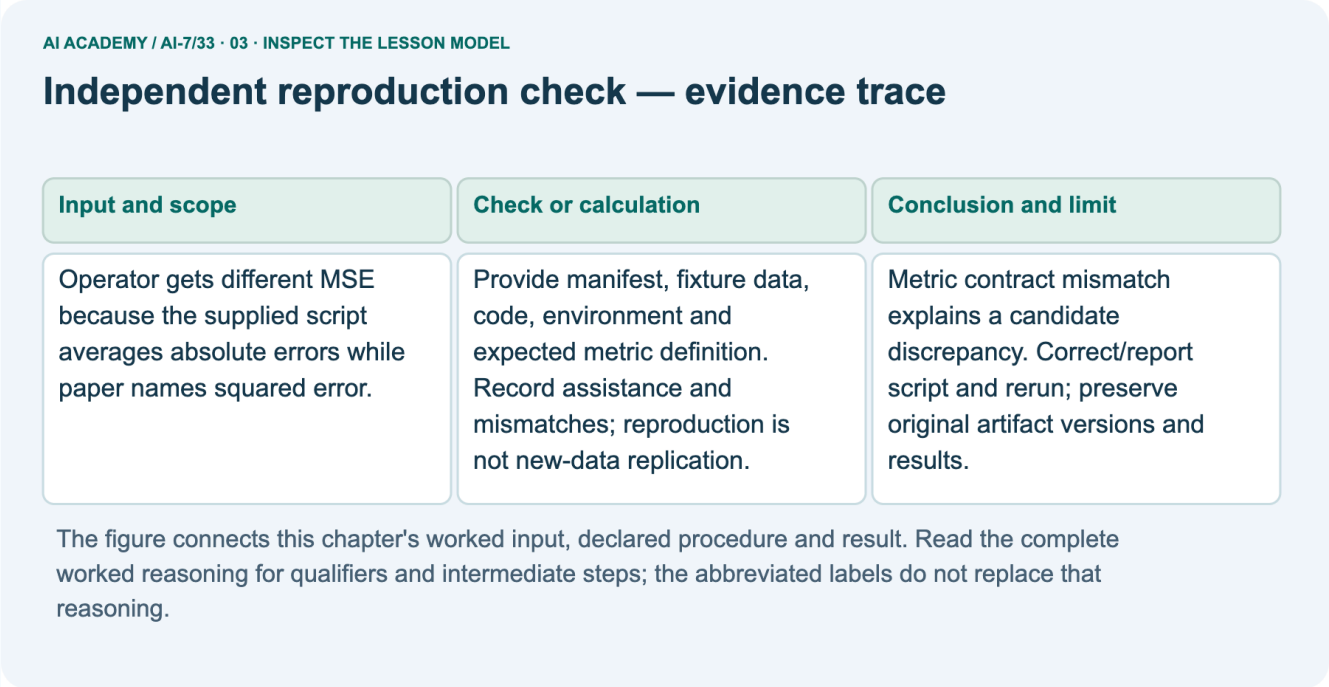
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare exact artefacts: Verify data and rubric versions before changing the result. → Locate disagreement: The score difference is concentrated in Q3. → Resolve by the rubric: Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Independent reproduction check — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Independent reproduction check — **evidence** trace.

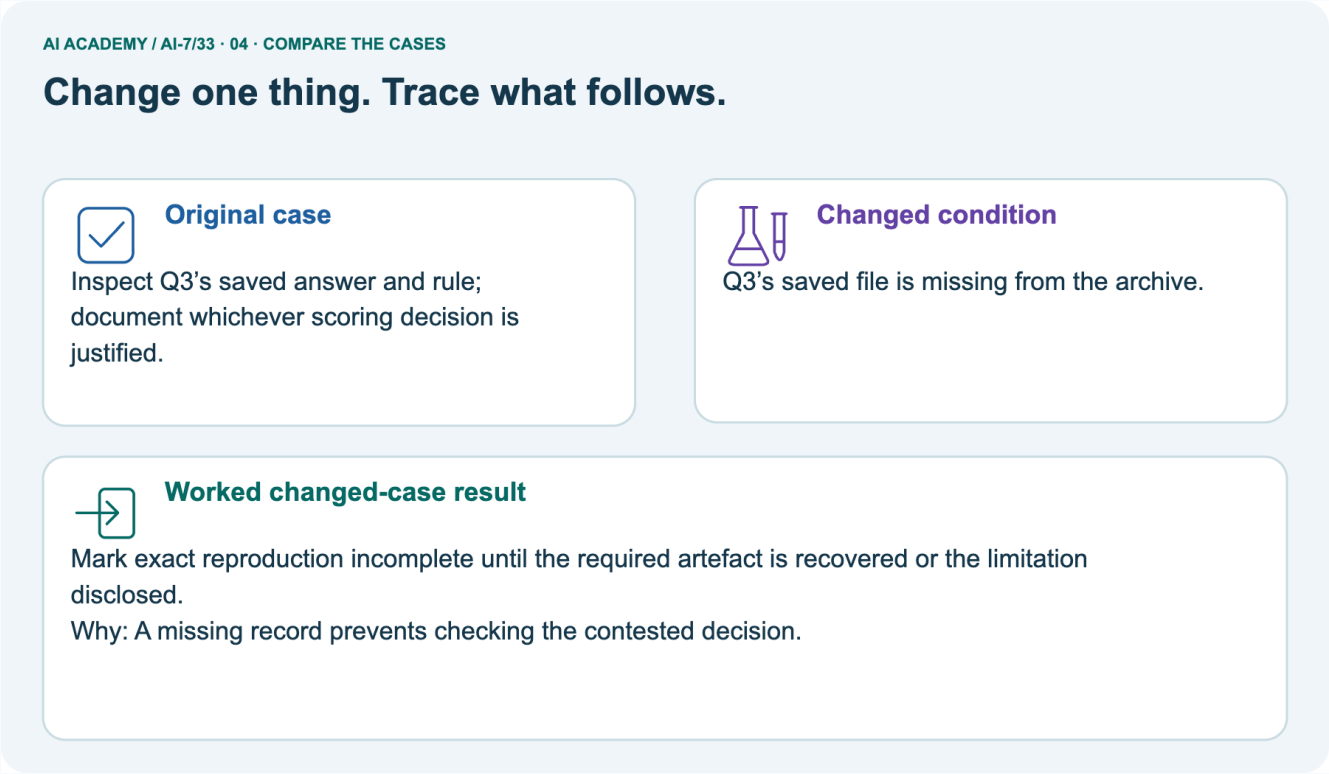
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. | Provide manifest, fixture data, code, environment and expected metric definition.

Record assistance and mismatches; reproduction is not new-data replication. | Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

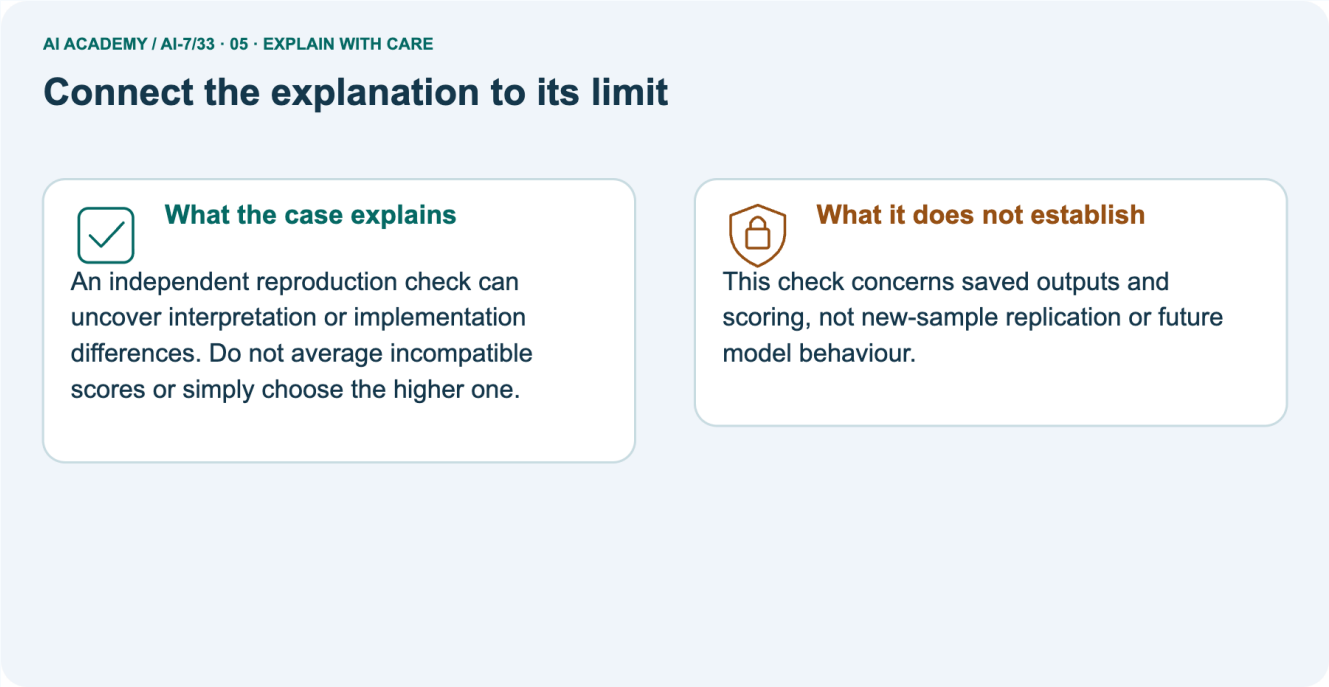
Original case: Inspect Q3's saved answer and rule; document whichever scoring decision is justified. Changed condition: Q3's saved file is missing from the archive.

Expected result: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. Reason: A missing record prevents checking the contested decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

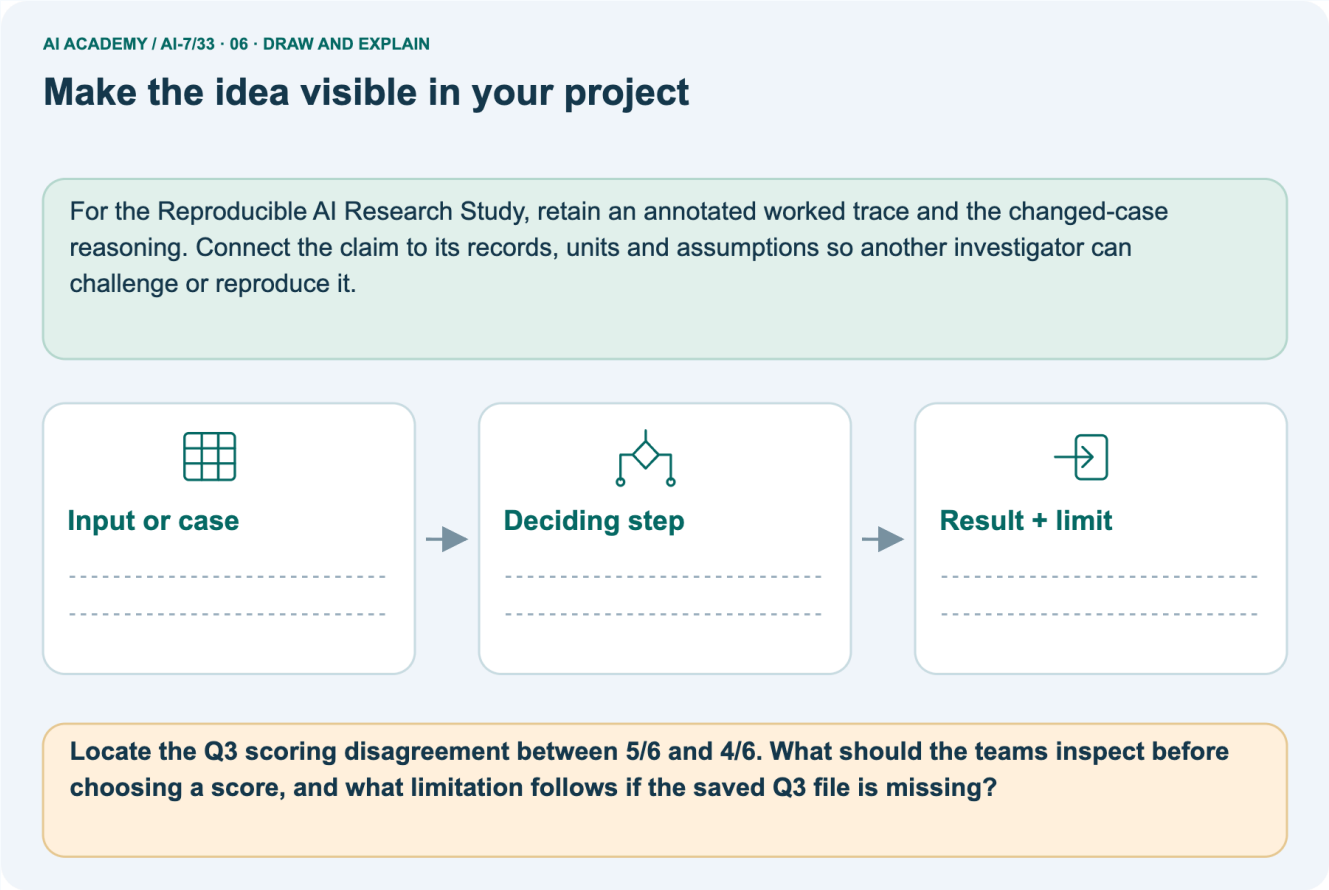
An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Limit: This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Independent reproduction check: the case

02. Follow the deciding steps

03. Independent reproduction check — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.